981674
Remigers Road
PUHOI

Rodney DISTRICT

No: 981674

WARKWORTH

COMBINED - PIM - BC

Sewage Con		100	Siting	190	Plumbing
Pans/New		110	Excavation	195	Drainage
Pans/Additional		120	Foundations	200	Ext. Cladding
Pans/Commercial		130	Bond Beams	205	Drainage
Urinals		140	Concrete Floors	215	Environ. Health
Water Connections		150	Pre-Line	240	Fire/Disabled Persons
Commercial		160	Roof Framing	250	Other
Private		170	Insulation	260	Final CCC
Dual Supply		180	Wall Lining		
Tank			Planned Inspections To Be Invoiced.		

Nil

Related PIM- BC No.: ..... Date Received: 20/7/98

Name: Nicklin

Address: Remigers Rd Puhoi

Granted Date - Actual: 10/8/98

Issue Date - Actual: 11/8/98.

Ward: Warkworth Lot: Allot 28

Details: Farm Shed/ Wc's-shop

D.P.: Parish of Puhoi

Passed to Hazlett.

Require certs for post wks and glue lam.

Pending letter Drafted: 21/7/98

Clerk to action PIM: 30/7/98

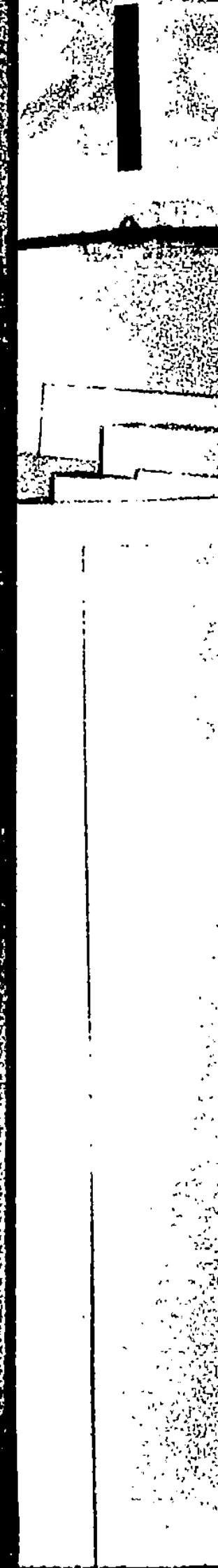
into rec 4.8 kc.

No	Action	Est	Actions Needed	Initial/Date Complete	Initial/Date Amended
002	Administration				
005	Technical Clerk Review	100		CRAD/7	
010	Check PIM to application	100			
015	Site Inspection - Building	100			
017	Site Inspection - P&D	200			
025	Development Engineer	400			
030	Planning	300		PKal 298	
035	Services Engineer	400			
040	Building Eng/ Struc/Geo	400			CAS 10/8.
045	Environmental Health	200			
055	Plumbing & Drainage	100			
060	Consultancy	400			
065	Check PIM/BC	100		6/8/98	
070	Grant PIM	500		7/8/98	
075	Check Building Consent	100	21. 25 31.7 95.	7/8/98	
080	Grant Consent	500		10/8/98	

Passed to LD

107

98 1 6 7 4



APPLICATION NUMBER 98/0674

OFFICER COMMENTS

Date	Comments
21. 7 98	General Rural.
30.7.98.	Give owner owners copy of Plans to take to Hutch. for Calls + Signature
28/7/98	Letter mailed to owner
7.8	Hutch received Kl.

FORM/DC008

10/05/96

PROJECT INFORMATION MEMORANDUM NO: ABA 981674

Section 31, Building Act 1991

ISSUED BY: RODNEY DISTRICT COUNCIL

APPLICANT	PROJECT
Name: NICKLIN, RAYMOND CHARLES Mailing Address: 49 REMIGER ROAD PUHOI 1240	THIS IS NOT A BUILDING CONSENT This is confirmation that after you have obtained a building consent and other necessary authorisations, the building work may then be undertaken.
PROJECT LOCATION	
Street Address: REMIGER ROAD, PUHOI, WARKWORTH Property Number: 5504019 Valuation Roll Number: Legal Description: CT 2100 PT ALLOT 28 PSH OF PUHOI	[ ] Building consent not yet applied for. [X] Building consent No: 981674 attached.
NEW CONSTRUCTION BUILDING (NOT DWELLINGS) Intended Use(s) in detail: SHED/WORKSHOP Intended Life: Indefinite, but not less than 50 years	[ ] Building consent applied for but not yet issued. [ ] Notification that the proposed building work may not be undertaken because a necessary authorisation has been refused.

This project information memorandum includes (cross each applicable box, attach relevant document, and send a copy to any relevant network utility operators and organisations having the power to classify land and buildings):

- [X] Information identifying relevant special features of the land concerned.
- [X] Information about the land or buildings concerned notified to the Council by any statutory organisation having the power to classify land or buildings
- [X] Details of relevant utility systems.
- [X] Details of authorisations which have been granted
- [X] Notification of any other authorisations which must be obtained before the proposed building work may be undertaken.
- [X] Details of authorisations which have been refused.
- [X] Details of relevant structural details.

Signed for and on behalf of the Council:

Name. [signature]

Date: 10/08/1998

NOTE: The Rodney District Plan is proposed to be altered by "Proposed Plan Change No. 62 - Introduction of New Financial Contribution Provisions". Contributions will be required in some situations where a building consent is applied for after the proposed plan change becomes operative. The proposed plan change is subject to the Resource Management Act 1991 submission and appeal process before it can become operative. A copy of the proposed change may be obtained from, and/or a copy of a booklet explaining the reasoning behind the proposed plan change may be purchased for $26.00 from the Council office at Orewa.

981674 NICKLIN

RODNEY ENVIRONMENT
BUILDING SECTION

PROJECT INFORMATION MEMORANDUM

This is not a building consent. A building consent must be issued before building work is started.

This is information from Council's records which is given as required by the Building Act 1991 for the building work proposed by application number 981674 to Rodney District Council dated 6/8/98 at Remiger Road, Puhoi,Warkworth for A Farm Shed.

Project Details

It is the responsibility of the applicant to show that the proposed building complies with the Building Act 1991.

Information Identifying Relevant Special Features of the Land Concerned

Nil

Information About The Land Or Buildings Concerned Notified To The Council By Any Statutory Organisation Having The Power To Classify Land Or Buildings

No details of classifications of land and buildings is known by the Council.

Details Of Relevant Utility Systems

Council has no record of any public utility systems on this land.

Details Of Authorisations Which Have Been Granted

Council has no record of existing authorisations for this project.

Notification Of Any Other Authorisations Which Must Be Obtained Before The Proposed Building Work May Be Undertaken

This activity is a permitted activity in terms of the Reviewed District Plan.
NB The building is not to be used for human habitation.

Details of Authorisations Which Have Been Refused

Council has no record of any authorisations for this project having been refused.

Project Confirmation

This is confirmation that the project may be undertaken subject to the requirements of the building consent and subject also to all other necessary authorisations being obtained. The project is subject to the provision of the Building Act 1991 and relevant documents.

(4)

BUILDING CONSENT NO: ABA 981674

Project Information Memorandum No: ABA 981674

Section 35, Building Act 1991
ISSUED BY: RODNEY DISTRICT COUNCIL

APPLICANT	PROJECT
Name: NICKLIN, RAYMOND CHARLES Mailing Address: 49 REMIGER ROAD, .." PUHOI 1240	All [X]
PROJECT LOCATION	Description: NEW CONSTRUCTION Nature: BUILDING (NOT DWELLINGS)
Street Address: REMIGER ROAD, PUHOI, WARKWORTH	Intended Use(s) in detail: SHED/WORKSHOP
LEGAL DESCRIPTION	Intended Life:
Property Number: 5504019 Valuation Roll No: Legal Description: CT 2100 PT ALLOT 26 PSH OF PUHOI	Indefinite, but not less than 50 years Estimated Value: $25000
COUNCIL CHARGES	Signed for and on behalf of the Council:
The balance of Council's charges payable on uplifting of this building consent, in accordance with the tax invoice are: Total: $ ALL FEES ARE GST INCLUSIVE	Name: ..... Date: 10/8/98

This building consent is a consent under the Building Act 1991 to undertake building work in accordance with the attached plans and specifications so as to comply with the provisions of the building code. It does not affect any duty or responsibility under any other Act nor permit any breach of any other Act.

This building consent is issued subject to the conditions specified in the attached pages headed "Conditions of Building Consent No "

Page 1.

981674 NICKLIN

CONDITIONS OF BUILDING CONSENT

SURFACE WATER

All surface water from roof, driveways and paved areas shall be collected and piped to an approved outfall.

TIMBER BARRIERS

All platforms higher than 1.0 metres above ground are to comply with F4 of the Building Code.

BUILDING CODE

All building work shall comply with the Building Code 1992, Acceptable Solutions and/or approved Alternative Solutions.

NZS 3604 - 1990

All building work to fully comply with NZS 3604 1990 Light Timber Frame Buildings.

ABA NO: 981674.

Date Inspected	
13-8-98 (1)	Footings Steel work as design. Trenches width + depth as design x 450 deep into firm clay sited in the middle of a paddock. Owner owns the land for at least 50m in all directions SNP
18.8.98 (2)	Slab: Polythene layer with D65 mesh. DPC Brickwork Rad. Floor thickening in place with 21Dia Bars as per plans SNP
26-9-98 (3)	Final: All building in place the following items requiring finishing prior to release of bond. Ground levels to top of Joists. (2) Unsightly brickwork to side of porch to be removed (Tenant aware will advise when completed) SNP
8 10 98 (4)	Final recheck - above two outstanding items completed

FORM/BG027 12/02/97

NO._____

CODE COMPLIANCE CERTIFICATE

Certifier	Signature	Date	Certifier	Signature	Date
BUILDING			POWER		
PLUMBING			GAS		
DRAINAGE			SERVICES		
E.H.O.			RESOURCE MGMT		
LIFTS			AIR COND.		
OTHER					

WORK IS NOW DEEMED TO BE COMPLETED. A CODE COMPLIANCE CERTIFICATE WILL BE ISSUED AS REQUIRED BY THE BUILDING ACT. CODE COMPLIANCE CERTIFICATE NO._____

Signature _____ Designation BFO 6 10 98 Date

WHEN THIS PROJECT IS COMPLETED THIS INSPECTION SHEET WILL BE ATTACHED TO THE RELEVANT PROPERTY FILE HELD AT THE COUNCIL OFFICE.

SPECIAL CONDITIONS:

CONSENT NO.	PIM NO.	LODGED AT SERVICE CENTRE	HEAD OFFICE DATE STAMP
981674			
DATE ISSUED	DATED ISSUED	DATE	

RECEIVED
20 JUL 1998
RODNEY DISTRICT COUNCIL

Rodney DISTRICT

Warkworth Service Centre, Baxter Street, Warkworth, Telephone (09) 425-8839
Hauraki Service Centre, Main Road, Hauraki, Telephone (09) 412-9128
Helensville Service Centre, Commercial Road, Helensville, Telephone (09) 420-8643

PRIVATE BAG 500, OREWA
TELEPHONE (09) 426-5149

4258539

BUILDING CONSENT APPLICATION

(Complete in all cases)

1. OWNER - As per attached Certificate of Title*

Name: R Nicklin
Postal Address: Remigant Rd. Puhoi
Phone Number: 4220800
Fax Number: 4220800
Previous Owner:

2. CONTACT (If not owner)

Contact Name:
Postal Address:
Phone Number:
Fax Number:

3. PROJECT LOCATION

Number:
Street/Road: Remigant Rd.
Locality: Puhoi
Ward: Warkworth

4. LEGAL DESCRIPTION

Valuation Number: 0092003800
Lot: PT LOT 28
DP: Puhoi PARISH
Certificate of Title:
Council Property ID: 5554019 37687 X7003025.
Site Area: square metres / hectares

5. PROJECT

New Building	✓
Alteration	
Relocation	
Demolition	

Description of Work: Shed Workshop
Intended Life: Indefinite but not less than 50 years or specified as ..... years
Estimated Value: $ 25000 (Incl GST) No of Units 1

SIGNED FOR AND ON BEHALF OF THE OWNER:
SIGNATURE: ..... DATE: 20-7-98

6. CHARGES — **OFFICE USE ONLY**

The Council's charges payable on the making of this application are:

$..... Receipt No: ..... Date: .....

This application will not be accepted until the Application Fee is paid.
Additional charges may be payable prior to uplifting of the Building Consent

* Copy of agreement for purchase from owner of title.

7.	Being stage ...1.... of an intended ...1.... stages.	

8.	FLOOR AREA OF PROPOSED WORK:	
	BUILDINGS OTHER THAN DETACHED ACCESSORY BUILDINGS:	
	Floor	Area square metres
	Basement	
	Ground Floor	130.
	First Floor	
	Second Floor	
	Additional Floors (Total)	
	Mezzanine	65.
	Decks	
	TOTAL	195
	DETACHED ACCESSORY BUILDINGS	Area square metres
	Garage	
	Carport	
	Other Buildings	
	TOTAL	

* Under Section 30 of the Building Act 1991, the applicant must be the owner of the land on which building work is contemplated or a person who or which has agreed in writing, whether conditionally or unconditionally, to purchase the land or any leasehold estate or interest in the land, or to take a lease of the land, while the agreement remains in force.

(Complete as far as possible in all cases)

9.

DESIGNER(S) Name:	Ruhoi Design
Address:	Saleyard Rd
Phone Number	Ruhoi Fax Number: 4220636
BUILDER Name:	Self.
Address:	
Phone Number:	Fax Number:
DRAINLAYER Name:	Reg. No.
Address:	
Phone Number:	Fax Number:
PLUMBER (if not as above) Name:	Reg. No.
Address:	
Phone Number	Fax Number:

CERTIFIER(S) (if applicable) Name:	Reg. No.
Address:	
Phone Number:	Fax Number:
Certifying:	
Name:	Reg. No.
Address:	
Phone Number:	Fax Number:
Certifying:	

PROJECT DETAILS

(Complete unless you have applied separately for a Project Information Memorandum)

10. The project involves the following matters. Tick each applicable box, if any, and attach the relevant information in duplicate.

(a) ☐ Location, in relation to legal boundaries, and external dimensions of new, relocated, or altered buildings.

(b) ☐ Details of any known or potential erosion, avulsion, falling debris, filled ground, subsidence, slippage, alluvion, inundation, hazardous contaminants on or near the site.

(c) ☐ Provisions to be made for vehicular access, including parking.

(d) ☐ Provisions to be made in building over or adjacent to any road or public place.

(e) ☐ Provisions to be made for disposing of stormwater and wastewater.

(f) ☐ Precautions to be taken where building work is to take place over existing drains or sewers or in close proximity to wells or water mains.

(g) ☐ New connections to public utilities i.e. gas, electricity, water supply, stormwater systems, wastewater systems.

(h) ☐ Provisions to be made in any demolition work for the protection of the public, suppression of dust, suppression of noise, disposal of debris and disconnection from public utilities.

(i) ☐ Details of any cultural or heritage significance of the building or building site, including whether it is on a marae.

(j) ☐ Details of any consents already obtained or approved relating to this application.

(Complete in all cases)

11. This application is accompanied by (tick each applicable box, attach relevant documents in duplicate).

(a) ☐ The drawings, specifications, and other documents according to which the building is proposed to be constructed to comply with the provisions of the New Zealand Building Code, with supporting comments, if any, including:

(b) ☐ Building certificates.

(c) ☐ Producer statements.

(d) ☐ References to accreditation certificates issued by the Building Industry Authority.

(e) ☐ References to determinations issued by the Building Industry Authority.

(f) ☐ Proposed procedures, if any, for inspection during construction.

NOTE

PRIVACY ACT 1993

(1) Personal information obtained in this application is primarily for establishing applicant identity and, when the application has been granted or declined, for administration, monitoring and enforcement.

(2) The information may be released to any inquirer about the application, the refusal of the application, or in relation to the use of any consent or approval of the application.

(3) The information will be held by Rodney District Council, 50 Centreway Road, Orewa.

(4) The information will be used for verifying and amending the Council's ownership and occupier records (the Rating Roll) and will be released in accordance with the Rating Powers Act 1988 and to Valuation New Zealand.

(5) You have the right to access the information and to request its correction.

NOTE The Rodney District Plan is proposed to be altered by "Proposed Plan Change No. 62 - Introduction of New Financial Contribution Provisions". Contributions will be required in some situations where a building consent is applied for after the proposed plan change becomes operative. The proposed plan change is subject to the Resource Management Act 1991 submission and appeal process before it can become operative.

A copy of the proposed change may be obtained from, and/or a copy of a booklet explaining the reasoning behind the proposed plan change may be purchased for $25.00 from the Council office at Orewa.

CODE COMPLIANCE CERTIFICATE NO: ABA 981674

Section 43(3), Building Act

ISSUED BY: RODNEY DISTRICT COUNCIL

BUILDING CONSENT NO: ABA 981674

PROJECT	PROJECT LOCATION
All [X] NEW CONSTRUCTION BUILDING (NOT DWELLINGS)	Name: NICKLIN, RAYMOND CHARLES Mailing Address: 49 REMIGER ROAD PUHOI 1240
Proposed Use(s) in detail: SHED/WORKSHOP Intended Life: Indefinite, but not less than 50 years	**LEGAL DESCRIPTION** Street Address: REMIGER ROAD, PUHOI, WARKWORTH Property Number: 5504019 Valuation Roll No: Legal Description: CT 2100 PT ALLOT 28 PSH OF PUHOI

This is:

[X] A final code compliance issued in respect of all of the building work under the above building consent.

[ ] An interim code compliance certificate in respect of part only, as specified in the attached particulars, of the building work under the above building consent. This certificate is issued subject to the conditions specified in the attached pages headed "Conditions of Code Compliance Certificate No:

Signed for and on behalf of the Council:

Name:.....Date: 06/10/1998

Rodney DISTRICT

Centreway Road, Orewa
Private Bag 500 Orewa

Telephone 0-9-426 5169
Facsimile 0-9-426 7280

770547

OFFICIAL RECEIPT

RECEIVED FROM: RAYMOND CHARLES NICKLIN

$ 983.00

Effective Date of Payment 20/ 7/1998

RAYMOND CHARLES NICKLIN
49 REMIGER ROAD
PUHOI

DETAILS		AMOUNT
BLDG 37687		983.00
Invoice 40835	ABA981674	983.00

OREWA STONE,T 20/ 7/1998 15:48:25 PAGE 1 of 1

BUILDING CONSENTS FEES

TAX INVOICE 40835
GST NO 10-683-092
DATE 20/07/1998

NICKLIN, RAYMOND CHARLES
49 REMIGER ROAD
PUHOI 1240

CONSENT APPLICATION NO: ABA 981674

OWNERS NAME:

PROJECT STREET ADDRESS: REMIGER ROAD PUHOI

CUSTOMER CODE: 37687

INVOICED TO DATE : 0.00

FEE CODE	DESCRIPTION	THIS INVOICE
002	Deposit Suspense	441.75
025	Branz	25.00
026	Bldg Consent Refundable Deposit	500.00
027	Building Industry Act Levy	16.25
	DUE & PAYABLE NOW :	983.00
	GST INCLUDED IN THIS INVOICE :	53.67

BUILDING CONSENTS FEES

CREDIT NOTE 42882
GST NO 10-683-092
DATE 19/10/1998

NICKLIN, RAYMOND CHARLES
49 REMIGER ROAD
PUHOI 1240

CONSENT APPLICATION NO: ABA 981674

OWNERS NAME:

PROJECT STREET ADDRESS: REMIGER ROAD PUHOI

CUSTOMER CODE: 37687

INVOICED TO DATE : 983.00

FEE CODE	DESCRIPTION	THIS INVOICE
002	Deposit Suspense	441.75 CR
008	Processing Charges Pre Issue	301.94
009	BC Admin Fee	39.37
011	A4 - Microfilming	12.60
012	A1-3 Microfilming	18.00
013	Actual Inspections	240.00
026	Blg Consent Refundable Deposit	500.00 CR
	REFUND DUE :	329.84 CR
	GST INCLUDED IN THIS INVOICE :	-18.91 CR

Rodney DISTRICT

DIRECT DOCUMENT PAYMENT

ACCOUNTS REF: _____ *DATE:* 20 October, 1998

Pay Refund To: RAYMOND NICKLIN
Mailing Address for Refund: 49 REMIGER ROAD
PUHOI

Property Owner: NICKLIN

Creditor Ref: ABA 981674

Document Date:	*Person I.D.:*	*Document Total: $*
20 October, 1998	37687	329.84

Original Receipt No.:	*Receipt Date:*	*Consent No.:*
770543	20/7/98	ABA 981674

Document Description: BLDG. REFUND - REMIGER ROAD

	Value $	*S.F.L Code*
1	329.84	OP9530-999-956
2		

_____
VCH/CHQ Checked

N J Lay
AUTHORISED

Rodney DISTRICT

28 July 1998

Reference ABA 981674
CB044

R Nicklin
49 Remiger Road
PUHOI 1240

Dear Sir

APPLICATION NO: 981674 FOR A FARM BUILDING
AT PT ALLOT 28 - ADDRESS: REMIGERS ROAD

With reference to your application for a building consent at the above site, the following information is required before the application can be (further) processed to completion:

1. Specific structural design calculations and details for the whole building. The NZS 3604 bracing calculations submitted are not applicable.

If you have any queries please contact Mr George Parnell at our Orewa Office, direct dial (09) 426 0216.

Yours faithfully

TECHNICAL OFFICER BUILDING
George Parnell

Telephone (09) 426 5169, Facsimile (09) 426 7260, DX BP 60009, E-mail mis@rodney.govt.nz
Correspondence to: Rodney District Council, Private Bag 500, Orewa, New Zealand
Head Office: 50 Centreway Road, Orewa, Service Centres: Warkworth, Helensville, Hospital

Land and Deeds Act
R.D.C. No. 7

NORTH

Reference: Vol. 757, Folio 202..
Transfer No
Order for N/C No. C.44703

NEW ZEALAND

Register-book
Vol 2100 Folio 23

CERTIFICATE OF TITLE UNDER LAND TRANSFER ACT

LIMITED AS TO PARCELS

This Certificate, dated the fifth day of November, one thousand nine hundred and sixty-two under the hand and seal of the District Land Registrar of the Land Registration District of AUCKLAND WITNESSETH that JOSEPH REMIGER of Puhoi, farmer

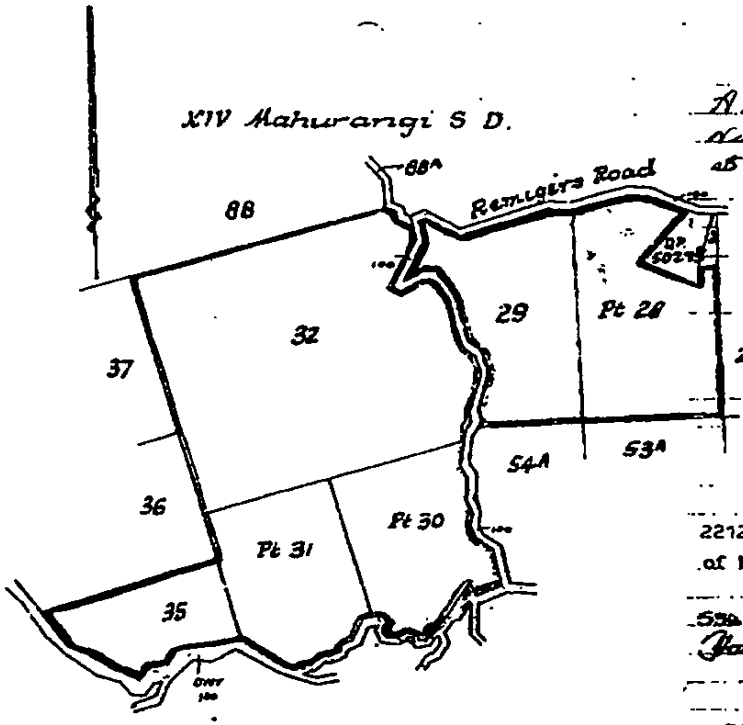
is seised of an estate in fee-simple (subject to such reservations, restrictions, encumbrances, liens, and interests as are notified by memorial under written or endorsed hereon, subject also to any existing right of the Crown to take and lay off roads under the provisions of any Act of the General Assembly of New Zealand) in the land hereinafter described, as the same is delineated by the plan hereon bordered Green, be the several admeasurements, a little more or less, that is to say: All that parcel of land containing 566 ACRES 1 ROOD 25.6 PERCHES more or less being Allotments 29, 32 and 35 and part Allotments 28, 30 and 31 Parish of Puhoi.

District Land Registrar

A150655 Transfer to Raymond Charles

23/01/98 17110 NZ LOANS ACBC/NORTHLAND + HARBOURTH

NO.712 1910



Total Area: 566:1: 25·6
Scale: 20 chains to an inch.

METRIC AREA IS
229.217960

Conversion Factors:
1 Acre = 4046m²
1 Perch = 25.29m²
1 Link = .2012 metres

A150655 Transfer to Reynard Hoole Nicklin of Sales person Produced 25.5.1966 at 10.20 a.m. and Entered 23.8.1966 at 11.00 a.m. A.L.R.

A150656 Mortgage to the State Advances Corporation of New Zealand Produced 25.5.1966 at 10.20 a.m. and Entered 23.8.1966 at 11.00 a.m. A.L.R.

A150657 Mortgage Remiger Produced 25.5.1966 at 10.20 a.m. and Entered 23.8.1966 at 11.00 a.m. A.L.R.

Variation of Terms of Mortgage A150656 produced 3/5/1968 at 2.05 o'clock.

A.L.R.

221241.2 Mortgage to State Advances Corporation of New Zealand 13.11.1975 at 11.55 o'c

for A.L.R.

539421.1 Caveat by Ronald Gordon Dunning of 275 Albany Road Browns Bay at 9.00 o'c

A.L.R.

813602.2 Mortgage to National Banking and Finance Corporation of New Zealand 6.11.1979 at 11.07 o'c

A.L.R.

864657.1 Variation of terms of mortgage 813602.2 - 17.9.1980 at 11.08 o'c

A.L.R.

23/01/98 17:10 BNZ LOANS AGRC/NORTHLAND + WARKWORTH NO.712 P011

Application No: 98/674

Site Address: Remigers Rd. Puhoi

Name: Nicklin.

REPORT ON PROJECT INFORMATION MEMORANDA /BUILDING CONSENT APPLICATION

Hazard Type	Report Site Specific	Report on Larger Area	Report on Adjacent Lot
Geotechnical			
Inundation			
Effluent			

OCK - Within Onerahi Chaos		Yes / No	
OCKS - Within Suspect Area OC		Yes / No	

362 - S36 (2) BA / S641A LGA			
AL - Alluvion			
AV - Avulsion			
CF - Certified Fill			
CM - Catchment Management Plan			
CNE - Consent Notice/Encumbrance			
EF - Effluent Disposal			
ER - Erosion			
FD - Falling Debris			
IU1 - Inundation - Flow Paths			
IU2 - Inundation - Flood Plains			
PC - Potentially Contaminated			
SP - Slippage			
SS - Subsidence			
TI - Technical Investigation Report			

Other Classifications	Report Site Specific	Report on Larger Area	Report on Adjacent Lot

Association Consulting Engineers of New Zealand

Institution Professional Engineers of New Zealand

JOB NO: Id5951

BCA No 981674

Building Regulation Clause (s)

PRODUCER STATEMENT - DESIGN

ISSUED BY: IAN THOMAS HUTCHINSON
(Suitably qualified Design Professional)

TO: RAY NICKLIN
(Owner)

IN RESPECT OF: PROPOSED DWELLING
(Description of Building Work)

AT: REMIGER ROAD
(Address)

LOT: Pt 25 DP: — SO: —

HUTCHINSON CONSULTANTS LIMITED (Design Firm) has been engaged by PUHOI DESIGN (Owner/Developer/Contractor)

to provide SPECIFIC STRUCTURAL DESIGN COMPUTATION (Extent of Engagement) services in respect of

the requirements of Clause(s) B1 (VM1) (VM4) (AS1) of the Building Regulations 1992 for

☐ All ☑ Part only as specified

of the building work. The design has been prepared in accordance with B1 (VM1) (verification method(s)/acceptable solution(s))

(respectively) of the approved documents issued by the Building Industry Authority and the work described on

PUHOI DESIGN (Design Firm) drawings title " PROPOSED SHED WORKSHOP AT REMIGER ROAD, PUHOI, FOR R NICKLIN

and numbered and the specification and other documents according to which the building is proposed to be constructed.

As an independent design professional covered by a current policy of Professional Indemnity Insurance to a minimum value of $200,000.00, I BELIEVE ON REASONABLE GROUNDS that subject to:

(i) the verification of the following design assumptions GOOD GROUND CONDITIONS IN ACCORDANCE WITH B1 (VM4)

and (ii) all proprietary products meeting the performance specification requirements, the drawings, specifications, and other documents according to which the building is proposed to be constructed comply with the relevant provisions of the building code.

Ian T. Hutchinson

Date: 06.08.1998

BE (CIVIL) M.E. MIPENZ
(Professional Qualification/s)

ERB Reg No. 9320

HUTCHINSON CONSULTANTS LTD
PO Box 150
OREWA
PH: (09) 426 6702 FAX: (09) 426 9669

Member ACENZ ☑ IPENZ ☑

This form to accompany Form 3 of the Building Regulations 1992 for the application of a Building Consent.

HUTCHINSON CONSULTANTS LIMITED
Civil & Structural Consulting Engineers
P.O. Box 150
Orewa
Auckland
Ground Floor Elan House
19 Tamariki Ave
Orewa
Telephone 0-9-426 5702
Fax 0-9-423 9569
AH 0-9-424 0013

DESIGN CALCULATIONS 981674

LETTER NO. 17122 AM
Rodney DISTRICT COUNCIL OREWA
- 7 AUG 1998
SUBJECT | FUNCTION | COPY
Buc | RLD |

The design contained herein remains the exclusive Copyright of Hutchinson Consultants Limited and shall not be reproduced in whole or in part by any means without their written authority.

CONTACT:
Puhoi Design
27 Saleyards Road
Ph 09 4220656
025 2215475

PROJECT:
R Nicklin
Proposed shed - workshop.
Remuera Road
PUHOI.

DATE:
4- August 1998

REFERENCE:
6120.

DESIGN BY:
Antony Cook.

Director: Ian T. Hutchinson M.E. (Civil) M.I.P.E.N.Z.

ACENZ

Design:

(A) Central Beam in dwelling $l = 3.6m$

roof

G

~~0.35 kPa x 2.5m = 0.89 kN/m (roof)~~

1.0 kN/m (external) = 1.0 kN/m (wall)

0.5kPa x 2.5m = 1.3 kN/m (floor)

$\Sigma G = 2.3$

Q

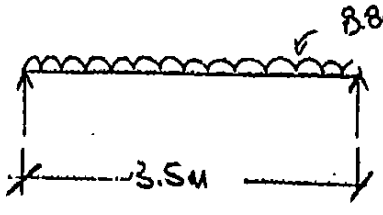
~~0.25 kPa x 2.5m = 0.63 kN/m (roof) (ψ_s) Q~~

1.5 kPa x 2.5m = 3.8 kN/m (floor) ($\psi_s = 0.7$) 2.7

$Q = 3.8$ kN/m $\quad \psi Q = 2.7$

(B) check (ULS)

$1.2(2.3) + 1.6(3.8) = 8.8$ kN/m



$M^* = wl^2/8$

$= 8.8 \times 3.6^2 \times 1/8$

$M^* = 14.3$ kNm

$M^* = 9.9$ kNm

timber member are too small, trial flitch plates.

$M^* = 14.3 \times 10^6$ kNm.

trial a
$\phi M_n = 153$ kNm trial a 300×100 $I_{xx} = 199.1 \times 10^6$ mm^4
trial a 250 × 10 flitch plate $I_{xx} = 13 \times 10^6$ mm^4
OR " " 200 × 12 " " $I_{xx} = 8 \times 10^6$ mm^4

C check SLS for Δs.

Floor ∴ $(G + \psi_s Q)$ = 2.3 + 2.7
w = 5.0 kN/m

$$I > \frac{wl^3}{38.4}$$

$$= \frac{5.0 \times 3.6^3}{38.4}$$

$$= 6.1 \times 10^6 \text{ mm}^4$$

250 × 10 flitch PL
with 2/300 × 50
(A)

OR. 300 × 100 timber member.

check SLS for Δs
(300 × 100)

⇒ (2 × 2.3) + 2.7
w = 7.3 kN/m

$$I > \frac{7.3 \times 3.6^3}{1.536}$$

$I > 128 \times 10^6$ mm^4
$I > 222 \times 10^6$ mm^4

Δ criteria fails.
for 300 × 100

Actual Δ

$$= \frac{5}{384} \times \frac{7.3 \times 3600^4}{8 \times 10^3 \times 199.1 \times 10^6}$$

= 10mm.

O.K if to acceptable
Δ is to be 10mm for Farm Shed
O.K?

(B) same loading as (A) ∴ same size pitch plates

Ruhoi

Design:

A) Central Beam in dwelling $l = 3.6m$

roof

G

~~0.35 kPa x 2.5m = 0.88 kN/m (roof)~~

1.0 kN/m (external) = 1.0 kN/m (wall)

0.5 kPa 2.5m = 1.3 kN/m (floor)

$\Sigma G = 2.3$

Q

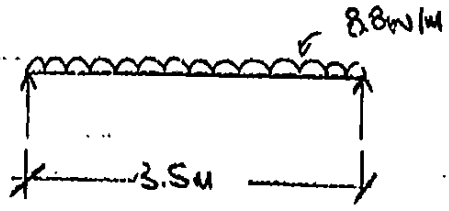
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$Q = 3.8$ kN/m $\quad \psi Q = 2.7$ kN/m

B) check (ULS)

$1.2(2.3) + 1.6(3.8) = 8.8$ kN/m



$M^* = wl^2/8$

$= 8.8 \times 3.5^2 \times 1/8$

$M^* = 14.3$ kNm

Beam Ⓒ, Beam over Porch $l = 4.0m$.

G

roof 0.35 kPa x 3.5m = 1.2 kN/m

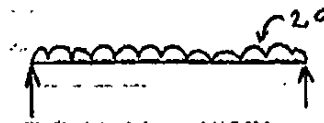
Q

roof 0.25 x 3.5m = 0.9 kN/m $\psi_l = 0$

$\psi Q = 0$

ULS

$1.2(1.2) + 1.6(0.9) = 2.9$ kN/m



$M^* = w l^2 \times 1/8$

$= 2.9 \times 4.0^2 \times 1/8$

$= 5.8$ kNm

trial 250 x 100 Beam.
$I_x = 113.8 \times 10^6$ mm^4

$\phi M_s = 10.6$ kNm $\therefore M^* < \phi M_s$

Check S.L.S.

timber $2G + 4Q = (2 \times 1.2)$
$= 2.4$ kN/m

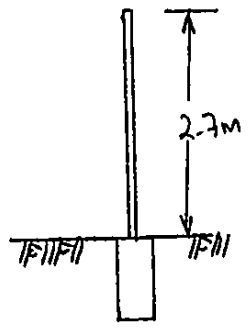
Roof Beam $\therefore I > \frac{2.4 \times 4.0^3}{2.048}$

$I > 75 \times 10^6$ mm^4 $\therefore$

250 x 100 OK ✓

Timber Posts:

100 x 100 H4 Posts.



$\phi M_n = 20.1\ MPa \times 0.85 \times 0.85 \times (325.52) \times 10^{-3}$ kNm

$\phi M_n = 20.1\ MPa \times 0.85 \times 1.0 \times (100 \times 100^2/6)$

$\phi M_n = 2.9$ kNm

$\therefore \frac{2.9\ kNm}{2.7\ m}$

= 1.1 kN force needed to fail timber post.

P = 1.1 kN

Rutledge formula

$A = \frac{1.17P}{SB}$

ϕ = 400mm for footings.
S = 100kPa
P = 1.1 kN

$A = \frac{1.17 \times 1.1}{100 \cdot 0.4}$

$A = 0.032$

$$H = A\left(1+\sqrt{1+2.18\frac{L}{A}}\right)+0.45$$

$$H = 0.032\left(1+\sqrt{1+2.18\left(\frac{2.7}{0.032}\right)}\right)+0.45$$

$H = 0.917m$

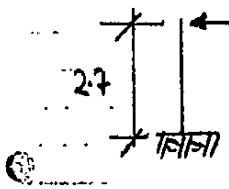
$\therefore$ $H = 1000mm$ deep.

Check capacity of 100mm piers

$q_{(z)} = 1.18$ kPa.

Area $= (3/2 * 3.2/2) = 2.4m^2$

Force $= 1.18\ kN/m^2 \times 2.4m^2 = 2.8kN$.



$M^* = 2.8 \times 2.7$

$M^* = 7.6kNm$

$(100\times100) = 98^2\times98/6$

$= 157\times10^3\ mm^3$

$\phi M_n = 0.80 \times 14.8 \times 157\times10^3 \times 1.0 \times 1.0$

$\frac{d}{b} = 1 \therefore 1^2-1=0$

$$S = 1.35\left[\frac{(2.7\times10^3)}{100}\left[(1)^2-1\right]^{0.5}\right]^{0.5}$$

$S=0 \therefore k_\theta = 1.0$

$\phi M_n = 1.87$ kNm

however A_{st} $\therefore$ $\phi M_n = 7.4$ kNm.

$\therefore$ needed are A x100 piers SED section thick. Psrt.

Calculations of Bracing Units.

$V(z) = V_{des} \times M_{(z\,cat)} \times M_s \times M_t \times M_r$.

$V = 49$

$M_{des} = 0.93$

$M_{(z\,cat)} = 0.93$ — Gawler. Terrain Category: ? ∴ 2

$M_s = 1.0$

$M_t = 1.16$?

0.91	1.0
5	10

$\frac{0.91-1}{5-10} = \frac{0.91-x}{5-6}$

$+ \quad 0.018 = -0.91 + x$

$0.93 = x$

$M_r = 0.9$

$V(z) = 49 \times 0.93 \times 0.93 \times 1.0 \times 1.16 \times 0.9$

$V(z) = 44$ m/s

$q(z) = (44)^2 \times 0.6 \times 1 \times 10^{-3}$

$q(z) = 1.18$ kPa.

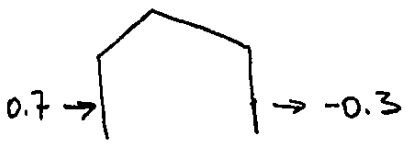
$F = \sum P_z A_z$

$P_z = (P_e - P_i)$

P_i - ignored

$P_e = C_{pe} \times K_a K_l K_d \times q_{(z)}$

C_{pe} =



$\alpha = 15°$

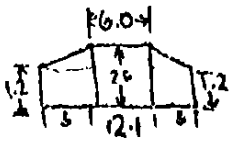
∴ $C_{pe} = 0.7 + 0.3 = 1.0$

$k_a = 1.0$
$k_e = 1.0$
$k_p = 1.0$ $\quad P_e = 1.0 \times 1.0 \times 1.0 \times 1.0 \times 1.18\ kPa$

$P_e = 1.18\ kPa$

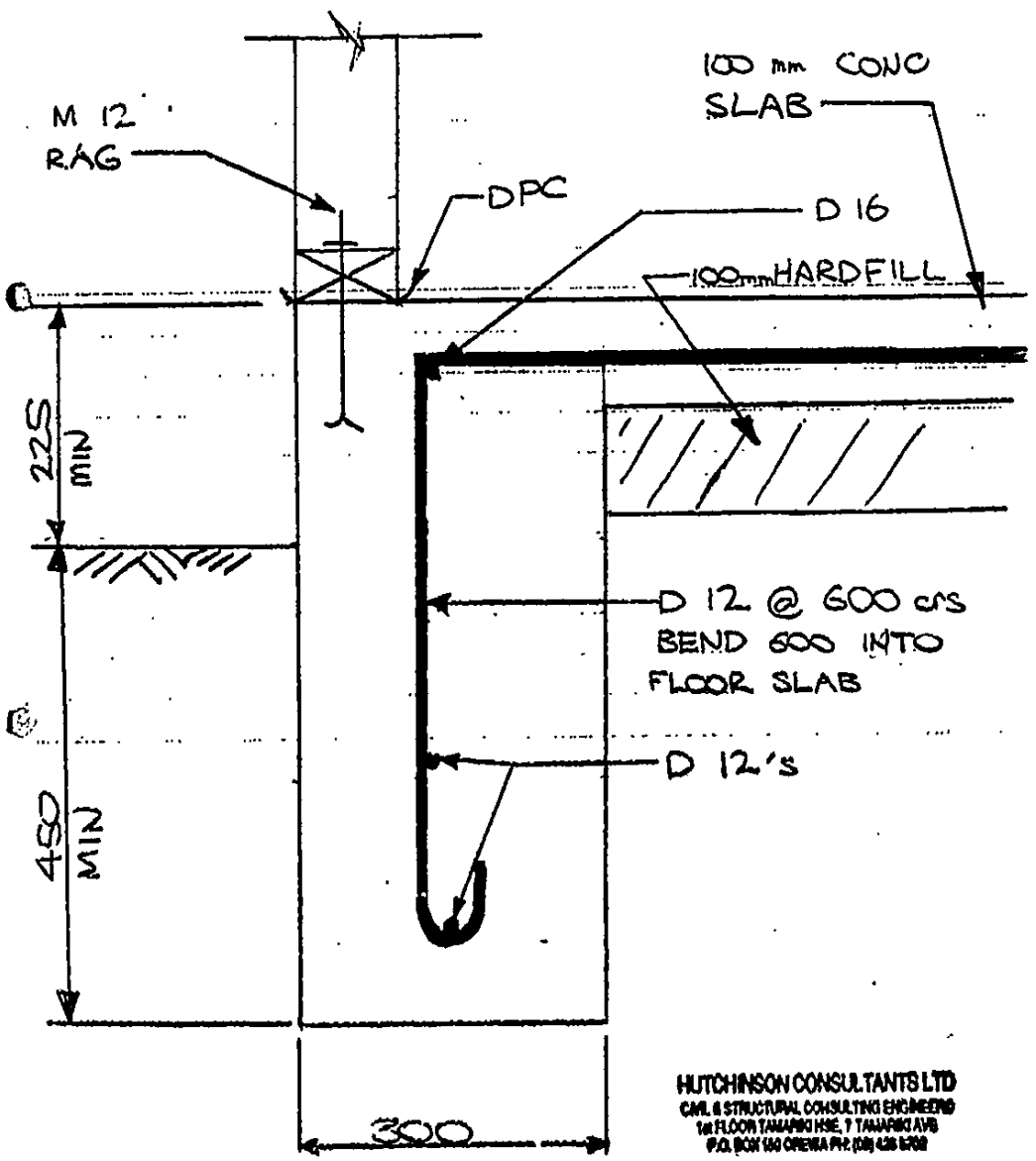
$D_z = 1.18\ kPa$

∴ North Elevation		Area	BU's
top	$(6 \times 1.8) =$	$10.8\ m^2$	260
Beton	(26.1)	$36.9\ m^2$	875

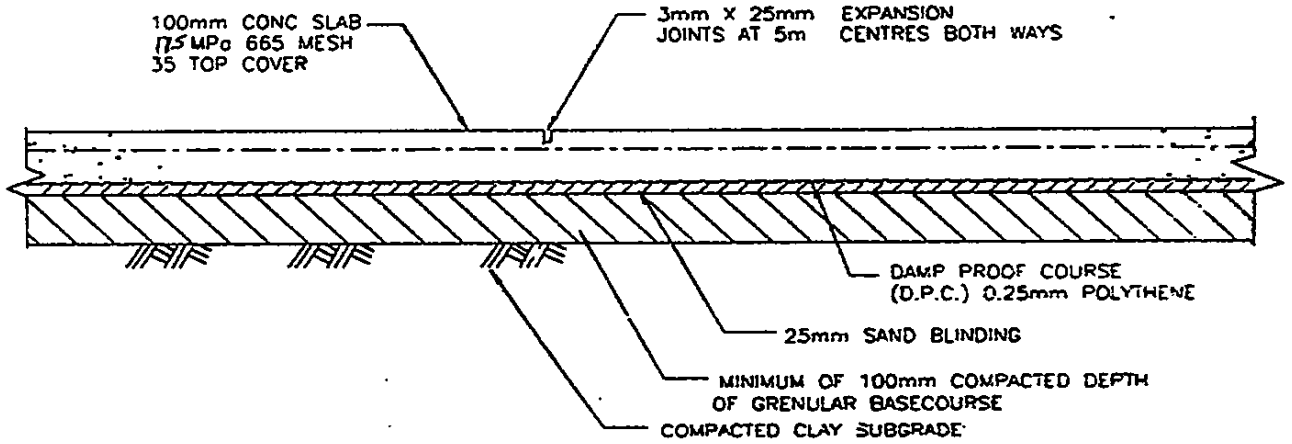


$\left[\frac{1.2+2.5}{2} \times 3 \times 2\right] + (2.5 \times 6.0) = 26.1\ m^2$

East	Area:	Bds.
top:	(11 x 1.8) = 198m²	470
Bottom:	(11 x 2.5) + 19.8 = 47.3	1120



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CIVIL & STRUCTURAL CONSULTING ENGINEERS
1st FLOOR TAMARIKI HSE, 7 TAMARIKI AVE
P.O. BOX 160 OREWA PH: (09) 426 6700



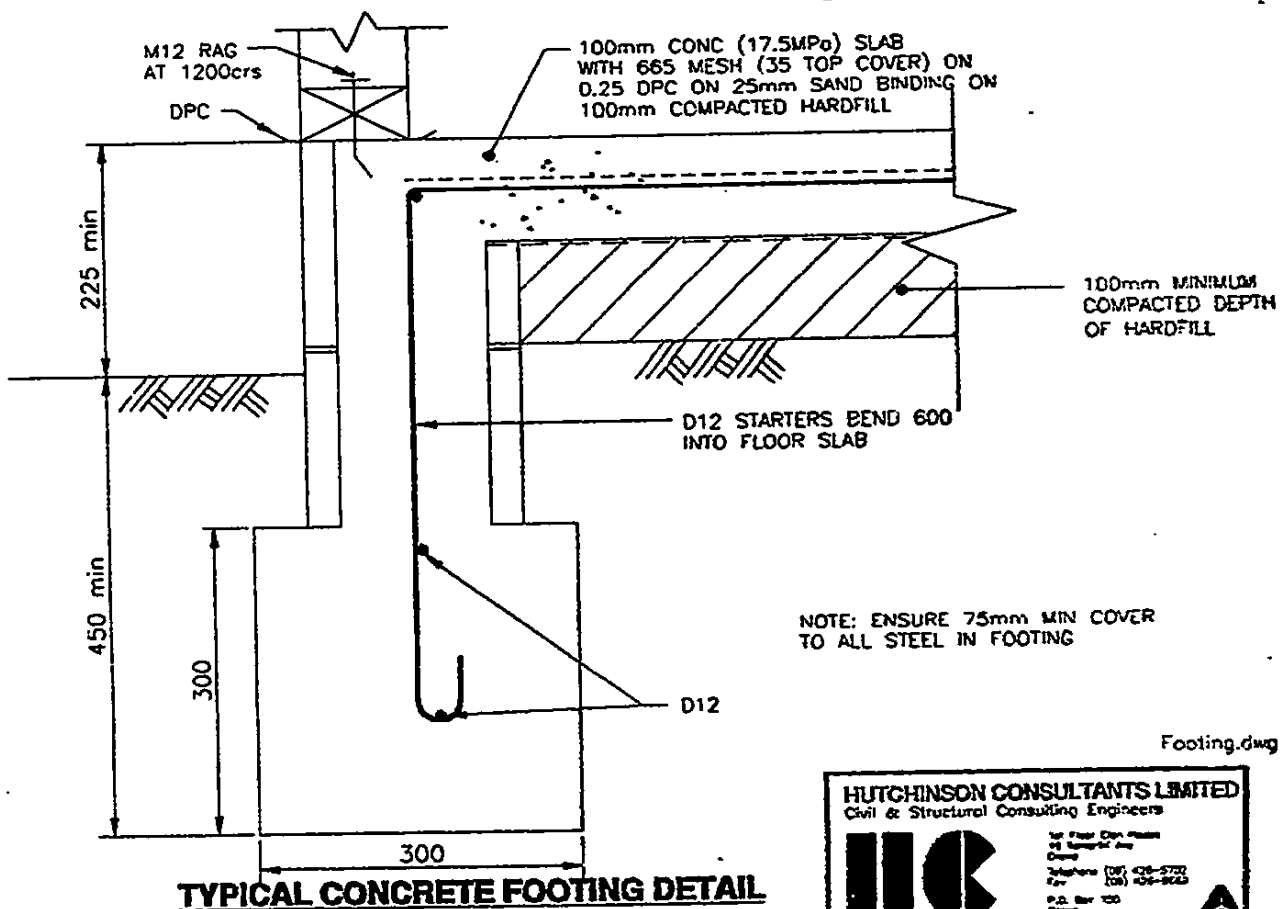
100mm 17.5MPa CONCRETE SLAB REINFORCED WITH 665 MESH
(3.5 TOPLAYER) ON 0.25mm THICK POLYTHENE D.P.C. ON 25mm SAND
LINDING ON 100mm MINIMUM COMPACTED DEPTH OF GRANULAR HARDFILL

Concslab.dwg

HUTCHINSON CONSULTANTS LIMITED
Civil & Structural Consulting Engineers

CONCRETE FLOOR SLAB TYPICAL DETAIL

SCALE 1 : 10



NOTE: ENSURE 75mm MIN COVER
TO ALL STEEL IN FOOTING

TYPICAL CONCRETE FOOTING DETAIL

SCALE 1:5

Footing.dwg

HUTCHINSON CONSULTANTS LIMITED
Civil & Structural Consulting Engineers

Workshop for Nicklin
Remuera Rd. 10 August '98 C. Shiu

300 x 100 beam 3m span.

DL	roof	0.3 x 3	= 0.9	LL	0.25 x 3 = 0.75
	wall		= 1.0		
	floor	0.5 x 3	= 1.5		1.5 x 3 = 4.5
	OW		= 0.3		
			3.7		5.25

1.2 DL + 1.6 LL = 4.4 + 8.4 = 12.8 KN/m

$M^* = 12.8 \times 3^2/8 = 14.4$ KNm.

$\phi M_b = 0.6 \times 17.9 \times 1.26 = 13.4$ KNm. (Marginal)

250 x 100 beam 3m span

Dr	deck	0.3 x 0.9	= 0.27	LL	1.5 x 0.9 = 1.4
	OW		= 0.3		
			0.6		1.4

Deck = 2 KN/m $M = 2 \times 9/8 = 2.2$ KNm (working)

Capacity 300 x 100 = 5.2 KNm (OK)

Please Photocopy

Wall Bracing Calculation Sheet A

Job Details — box 1

Name R NICKLIN

Street and Number

Lot and DP Number

City/Town/District

Location of Storey: ~~single/upper of two/lower of two~~

Building height to apex	6 m	Roof weight	~~light/heavy~~
Roof height above eaves	3 m	Cladding weight	~~light/heavy~~
Stud height	3 m	Room in roof space	y/~~n~~
Average roof pitch	15°		
Building length BL =	12 m	Gross Building	
Building width BW =	10.9 m	Plan Area,	GPA = 130 m2

Note: When the average roof pitch is over 25 degress, use the eaves length and width to determine BL and BW.
Note: For heavy roofs use the roof plan at eaves level to determine GPA.

Wind Zone — box 2

Region:	Terrain:	Exposure:	Topography:
R1 0	Inland 0 ✓	Sheltered 0	Gentle 0 ✓
R2 1 ✓	Coastal 1	Exposed 1 ✓	Moderate 1
			Extreme 3

Total points 2

Wind zone: Low (0) — Very high (3)
Medium (1) — Specific Design (4)
✓ High (2)

Earthquake zone — box 3

From figure EQ1 select Earthquake Zone: A B (C)

BUs required Wind — box 4

From Table W1A/W1B
W along = 87 BUs/m
W across = 87 BUs/m
Total wind load,
W ALONG:
W along x BW = 948 BUs
W ACROSS
W across x BL = 1044 BUs

BUs required Earthquake — box 5

From Table EQ1
E = 2.4 BUs/m2
Note: For a room in the roof space use E+1
Total earthquake load,
EQ ALONG and EQ ACROSS:
E x GPA BUs = 312 BUs

GIB

Please Photocopy

Wall Bracing Calculation Sheet B

Along

Wall or Bracing Line		Bracing Elements Provided			Wind		Earthquake	
1	2	3	4	5	6 W	7 W	6 E	7 E
Line Label	Minimum BUs Required	Bracing Element No.	Bracing Type	Length Element (m) L	Rating BU/m W	BUs Acheved (BU/m x L) W	Rating BU/m E	BUs Achieved (BU/m x L) E
A	120	A1	SP1	1·2	100	120		
		A2	SP1	3·0	100	300		
B	70	B3	SP1	0·9	100	90		
C	70	C4	SP1	0·9	100	90		
D	120	D6	SP1	3·6	100	360		
E								
Totals Achieved					W	960	E	
From Sheet A Totals Required					W	948	E	912
Wreq/Ereq =								

*If Wreq/Ereq is 1 or less complete E column only
If Wreq/Ereq is 1.5 or more complete W column only
Otherwise complete both W and E

Across

Wall or Bracing Line		Bracing Elements Provided			Wind		Earthquake	
1	2	3	4	5	6 W	7 W	6 E	7 E
Line Label	Minimum BUs Required	Bracing Element No.	Bracing Type	Length Element (m) L	Rating BU/m W	BUs Achieved (BU/m x L) W	Rating BU/m E	BUs Achieved (BU/m x L) E
M	108	M6	SP1	3·6	100	360		
N	70	N7	SP1	3·6	100	360		
O	108	O8	SP1	3·6	100	360		
P								
Q								
Totals Achieved					W	1080	E	
From Sheet A Totals Required					W	1044	E	312
Wreq/Ereq =								

GIB

SPECIFICATION

For the erection of SHED & WORKSHOP

at RENIGER RD PUHOI

for B NICKLIN

This is the specification referred to in an "Agreement" (1) between

..... as Owner, and

..... as Builder,

dated ..... and if not mentioned the Agreement (1) takes notice of the following completion and penalty clauses.
Time for completion ..... weeks from acceptance of tender.
Liquidated damages for delayed completion shall be $..... per week. Extensions to the contract period for extra work or for contingencies beyond the control of the contractor to be as agreed between the parties at the time and added to the time for completion.(2)

*PAYMENTS AND BONDS:

Standard monthly progress claims that allow ready assessment of the quality and value of the work carried out shall, subject to compliance with contract requirements, be processed within 10 days of receipt.
For purposes of ensuring adequate and fair performance throughout, catering for a 30 day Maintenance (Defects Liability) period as under P & G No. 10, 10% of the contract sum shall be retained as a form of Contractor's Bond in accord with:-

(a) Addendum D of the N.Z.I.A. General Conditions of Contract
(b) Form F.H.4 from N.Z.M.B.F., or
(c) As agreed between the owner's lawyer and the builder's lawyer.

ARBITRATION:

If dispute or difference should arise between the Owner and Contractor the Contract shall be subject to arbitration under the Arbitration Act 1908.

*Foot Notes:

(1) The Agreement referred to and a suitable form of contract (Conditions of Contract) shall be provided by the Owner ..... or Contractor ..... and is to be completed by all parties before the site work is commenced.

(2) All additions and deletions made in this Specification before signing of the Agreement shall be dated not later than the date of Agreement. Any subsequent additions, omissions or variations to the contract shall be authorised in writing and an agreed price stated.

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CARPENTER AND JOINER ..... PAGE NO. 10
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DRAINLAYER ..... PAGE NO. 23
SOLID PLASTERER ..... PAGE NO. 25
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PAINTER AND GLAZIER ..... PAGE NO. 28

IT IS IMPORTANT TO READ "LIMITATIONS" AND "GUIDANCE" PRINTED BELOW. CURRENT COPIES OF THE NEW ZEALAND STANDARDS MENTIONED, PARTICULARLY IN CLAUSE 1 OF "PRELIMINARY AND GENERAL" (PAGE 2), CAN BE PURCHASED FROM THE AUCKLAND BUILDING CENTRE AND SHOULD BE KEPT ON SITE.

LIMITATIONS

This specification is essentially for a house to be built in accordance with NZS 3604 : Code of Practice for "LIGHT TIMBER FRAME BUILDINGS not requiring specific design...." otherwise referred to simply as NZS 3604.

The code allows houses on timber or concrete pile system, pole platform foundation or on foundation walls of reinforced concrete or masonry. The latter foundation should be in accord with NZS 4229 : "Code of Practice for concrete masonry buildings not requiring specific design".

NZS 3604 provides for so many variables, via new techniques and materials and its greater engineering approach, that a short standard specification of this type has to be limited to a "Guidance Only Format".

To use THIS specification for building permit application purposes, and for accurate pricing, it becomes more important than before to DELETE non-applicable clauses and to ADD new or more appropriate clauses - according to the Owner's, designer's and builder's greater freedom of choice.

The necessity to delete and add will increase the more the proposed new house is to differ from the normal 2.4m stud, single storey type.

GUIDANCE

Please do not neglect to make the necessary deletions and additions to this specification. Also ensure that your drawings for the house are correct according to the deletions and additions made.

The Auckland Building Exhibition Centre contains a wide range of ideas and products available on the market for inclusion in your design. Please enquire for further details from the helpful staff on Reception, or mail the 'Information Request' form included in this specification direct to P O Box 271, Auckland 1. We also have available by appointment Independent Technical Building Consultants who, for a consultation fee, are available to advise on Building, Construction, Engineering and Architecture.

Revised Edition 1991 by: Able Building Consultants, P O Box 36-165, Northcote, Auckland. Ph: 418-3868

PRELIMINARY AND GENERAL

1. **SCOPE:**
This specification conforming with the attached Agreement and Conditions of Contract describes the work to be done and the materials to be used in the construction of the house shown on the attached drawings. The Contractor is expected to be familiar with the site and the Local Building Bylaws, especially where relevant clauses of NZS 3604 and NZS 4229 apply to the contract. The above number NZ Standards shall be included in the contract documents and kept on site and made available as necessary for purposes of supervision by the owner or his agent to ensure compliance of work during and on completion of the contract.

2. **PERMITS:**
Contractor to comply with the Labour and Building bylaws of the district, to apply for an obtain all the necessary permits and to pay all fees for same, unless otherwise mentioned.

3. **CONSTRUCTION ACT:**
The provision of the Construction Act 1959 and regulations are deemed to be part of this contract.

4. **PROVIDE & FIX:**
The words "provide" and "fix" shall be construed to mean "provide" and "fix" where mentioned separately unless otherwise mentioned.

5. **INSURANCE/INDEMNITIES:**
The Contractor to have all of his employees covered against accident by an Employers' Liability Policy and to take out insurance against fire and theft of materials off-site for a sum sufficient to cover the full amount of the Contract Sum, both policies to remain in force until the building is taken over by the Owner; the fire insurance to be in the joint names of the Owner and Contractor.

* The Conditions of Contract shall be approved, or include "Special Conditions of Contract" issued by the Building Performance Guarantee Corporation as necessary to effect a formal indemnity agreement.

6. **INTERPRETATION:**
Work or materials shown on the drawings or specified and not shown, must be supplied as though both shown and specified. Materials shown but not specified must be of the kinds commonly employed for the service they are intended to perform. All figured dimensions shall be taken in preference to scale. The Contractor shall be held responsible for the setting out of work and he shall make good at his own expense any errors that occur through his lack of checking or faulty workmanship.

7. **STABILITY:**
The Contractor shall carefully brace and support all parts of his work against damage by wind and protect work from the elements as necessary during construction.

8. **PROTECTION OF PROPERTY:**
The Contractor shall protect adjoining properties during the contract and shall make good all damage at his own expense.

9. **DAMAGE:**
The Contractor shall make good at his own expense and to the satisfaction of controlling authorities, any damage done to footpaths, kerbs, drainage, etc., or other property under control of such authorities. Each trade shall take care to prevent damage or disfigurement of the work of other trades and will be responsible for cost of restoring same.

* THIS IS A WISE PRECAUTION

10. **MAINTENANCE:**
Owner not familiar with building procedures and standards, are advised to appoint an independant consultant to undertake an inspection and compile a list of matters that may require attention at the commencement of the 30 day period.

11. **MATERIALS:**
Any materials herein specified that are not procurable at the time they are required, thus tending to retard the progress of the Contract, may be substituted with other similar materials, providing that the substituted materials conform to the Local Bylaws and with permission of the Owner. The Contractor is first to notify the Owner of any change proposed and at the completion of the Contract will adjust any difference in cost.

12. **CONTINGENCIES:**
Provision is made by the Builder and Owner to meet any contingencies that may arise due to the fluctuations in the price of various materials or labour. Should there be either a rise or fall in the price of labour or materials, from the date that the tender is submitted until final payment, an adjustment to the Contract Price, is to be made accordingly provided that the Contract Price has been affected by such rise or fall in prices. The sum of $ is allowed in the Contract Price as a Contingency Fund. The Owner will expect the Builder to submit proof of any increases claimed for by way of invoices or labour costs.

13. **VISIT SITE:**
Tenderers shall visit the site and ascertain the nature and extent of the work and the rights and interests that may be interfered with and any other matter that may influence the making up of a tender or the carrying out of the Contract in its entirety. The levels shown on the drawings are approximately correct but tenderers shall verify these as no claims for extra will be allowed on the basis of incorrect levels shown.

14. **WORKMANSHIP:**
All work shall be carried out in accordance with the best trade practice, in strict conformity with the drawings and specification to the satisfaction of the Owner. All defective or damaged work shall be removed and made good to the satisfaction of the Owner.

15. **CLEANING:**
The Contractor, at the conclusion of the Contract, shall have all ceilings, walls and woodwork carefully dusted and wiped down, windows washed and glass free from scratches, floors brushed and wiped down and the entire building left in a perfectly clean condition for occupation.

EXCAVATOR

1. **PRELIMINARY AND GENERAL:**
Note where Preliminary and General clauses apply to this trade.

2. **ALLOWANCE IN TENDERING:**
Allow for foundation depths as shown on drawings (450mm below cleared gound level as if in an expansive clay). Any variations from dimensions shown to be adjusted in terms of Clause (Insert Clause No.) of the General Conditions of Contract.

3. **SETTING OUT:**
Set out the work as shown or implied on the drawings. Check accuracy in terms of position, levels and square.

CONTINUED...

EXCAVATOR - CONTINUED

4. CLEAR SITE:
Remove vegetation, trees, roots and 150mm top soil* within area to be covered by the building, driveway, paths, terraces and steps. Do not remove any other trees without Owner's consent.
* Note: Alternatively with consent of Owner and Local Building Authority, leave topsoil in place where excavation would otherwise cause ponding under timber ground floors (see Clause 3.4.1 of NZS 3604). Deposit top soil in heap as directed and avoid covering with subsoil subsequently excavated.

5. LEVELLING AND BULK EXCAVATION:
Excavate for all site levelling, foundation walls and/or piles, underground services and subsoil drains etc. to correct levels, to firm bearing or to obtain sufficient frictional resistance* to satisfy Local Building Authority.
* Note: Item 5 allows for driven timber piles to NZS 3605.

6. FOUNDATION TRENCHES:
Take out trenches straight, level and to proper width and keep free of water and loose material. Concrete shall not be placed until excavations approved. If, other than under item 2 of this section, trenches are dug too deeply then such excess depth shall be filled with concrete, as specified for foundations, at the Contractor's expense.

7. HARDFILL:
Use only approved fill not less than 100mm thick and well compacted under concrete ground floors and wherein otherwise specified. Blind surface of fill with not less than 5mm of sand.

8. SITE DRAINAGE:
Take out land drainage trenches to adequate depth and falls to prevent dampness under the building. Provide for field tile drains (or perforated plastic drainage pipe) bedded on scoria. Back fill with clean graded scoria from 50mm to fine.

9. HOUSEHOLD DRAINAGE:
See Drainlayer Section.

10. BACKFILL AND TOPSOIL:
Backfill and well consolidate in 100mm layers to the foundation walls, pile footings and service trenches. Spread previously excavated top soil as directed. Do not damage any waterproof coatings or polythene protecting foundation walls from ground water entry.

CONCRETE AND REINFORCING

1. PRELIMINARY AND GENERAL:
Note all clauses under Preliminary and General of this specification which shall apply to this section of the work.

2. EXTENT OF WORK:
Comprises the setting out, boxing and placing of concrete in the foundations, floor slabs, walls, beams and bands and any other concrete work shown in the drawings.

3. MATERIALS:
Concrete which shall be ordinary grade and comply with requirements of NZS 3109 Builders mix may be used if agreed by Owner, providing the minimum crushing strength is 17.5MPa at 28 days standard cure is unaffected.

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4. FORMWORK:
Formwork shall be so constructed that the concrete thickness and shapes required are obtained as detailed without removal causing damage. Times of removal of formwork, methods of construction and pouring and curing of concrete shall be as set down in NZS 3109.

5. CONCRETE FOUNDATION WALLS:
Foundation footings, walls and reinforcing shall be to the sizes shown on drawings in accordance with Table 4 and relevant clauses of NZS 3604, supporting single storey, 2 or 3 storey as appropriate. Such walls shall finish not less than 225mm above finished ground level but not measure more than 2m above the bottom of footings.
NOTE: A full (2.4m) storey concrete (or concrete masonry) foundation wall is outside the scope of NZS 3604 but is permissible either by:-
(a) Construction to NZS 4229 for example using concrete block masonry,

or
(b) Specific Design using concrete or masonry.
It is then permissible to erect one or more storeys to NZS 3604 on top of such foundation walls.
Reinforcing steel shall be lapped at least 40 rod diameters for plain rods and not less than 30 diameters for deformed bars in concrete. (See Concrete Blocklayer for other laps).

6. CONCRETE PILES:
Ordinary concrete piles to NZS 3604 shall for this project be precast piles embedded into insitu footings. All other concrete piles, ie: anchor piles, braced piles and cantilever piles shall be formed insitu where indicated on drawings, reinforced and to depths in accord with NZS 3604.

7. CONCRETE FOOTINGS FOR EMBEDDED TIMBER PILES:
The minimum plan dimensions of square or circular footings, thickness of pile footings and embedment of the respective timber piles into such footings shall all accord with NZS 3604 for ordinary, anchor, braced and cantilever piles with any ends cut on site uppermost and brush treated according to Timber Preservation Authority requirements.

8. CONCRETE SLAB-ON-GROUND FLOORS:
Floors having foundation walls that support only one storey need not be reinforced providing that shrinkage control joints are provided at 3m spacings for 75mm thick slabs or 4m spacings for 100mm thick slabs. In all other cases concrete slab-on-ground floors and porches shall not be less than 100mm thick on well compacted granular fill and reinforced with a layer of 668 welded reinforcing mesh lapped 225mm at joins (or 665 mesh if cast in more than 15mm maximum dimension in one operation). Reinforced slab-on-ground proches may be cast separately from foundation walls but shall be securely anchored to them with R6 bars at 600mm c/cs lapped not less than 300mm with the slab reinforcing.
Where floor slabs occur in rooms for habitation they shall have 0.25mm thick polythene sheet under, either welded into one continuous sheet or with joints lapped not less than 150mm and sealed with a self adhesive tape to manufacturers' specification. The Contractor shall not use any other vapour barrier without approval in writing and shall protect from punctures and tears before and during placement of concrete. Any damage done to the vapour barrier during provsion of underslab or in the slab services shall also be made good before placing concrete.

9. SUSPENDED CONCRETE FLOORS:
Precast, prestressed suspended concrete floor deckings shall be those manufactured by

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All junctions with external walls shall included adequate tying in as per manufacturer's detail, and where there is brick veneer above provide for a solid minimum 50mm deep step detail treated with a bitumen water proofing compound to ensure freedom from rainwater entry. All insitu concrete placed over units and at ends to be of strength at standard 28 days cure prescribed by the manufacturer. Where metal trough sections supporting concrete floors are shown erect sections and place concrete all as details and instructions provided by.....

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10. BUILD-IN BOLTS, ETC:
Provide in concrete for openings for vents or as required by other trades and for holding down bolts in accordance with NZS 3604. Timber grounds where required for fixing door frames etc. shall be heart totara dovetailed and where required water bars shall be of brass or galvanised iron. Powder powered tool fixings can be used where appropriate.

11. PATHS:
Where shown on drawings paths shall be laid not less than 75mm thick reinforced with 668 mesh. The surface shall be graded evenly to provide suitable drainage and left with a uniform finish using a wooden float.

12. COMPLETION:
Leave all clean and tidy at finish and made good any defective work.

CONCRETE BLOCKLAYER

1. PRELIMINARY AND GENERAL:
Read and note all clauses under Preliminary and General of this specification where they apply to this trade.

2. EXTENT OF WORK:
Refer to drawings for layout and extent of work. Build the whole of the reinforced and unreinforced 200mm concrete blockwork. Build in as the work proceeds all reinforcing steel, bolts and all other lugs, conduits, sleeves and openings for vents etc., required for the work of other trades. Keep surface clean and free from mortar, perpends true and faces true to line and vertical. Blocks are to be kept dry before laying. Reject all blocks not complying with NZS 3102. Reinforced masonry foundation walls not exceeding 2m in height from the underside of the footings shall be to NZS 3604 or to NZS 4229 (specify preference). Where exceeding 2m in height construct in accord with NZS 4229 or to specific design.

3. BOND:
Blocks shall be laid in stretcher or stack bond as shown. All joints shall be full and struck to provide a fair faced finish to both faces. Complete bond shall be secured between the blocks and mortar. Ensure that bond is not broken by making adjustments to blocks after mortar has taken a set.

4. MORTAR:
The mortar for all blockwork shall be composed and mixed according to the relevant NZS clauses; water shall be of drinking quality; sand shall comply to NZS 3109 for grout filling of masonry cavities and NZS 3103 "Sands for Mortar, Plaster & External Renderings" for mortars.
All cement shall comply with NZS 3122 and shall be properly stored at the site and adequately protected from dampness.

5. MINIMUM REINFORCING STEEL REQUIREMENTS:
Minimum reinforcing steel requirements in concrete masonry on this contract shall be relative to the height of the masonry, accord with the following:-
(a) Where shown less than 2m in height from underside of footings to top of foundation wall - to NZS 3604.
(b) Where exceeding (a) above, where a full 2.4m or higher storey height is shown, be:-
(i) to NZS 4229
or
(ii) to Specific Design; and work in this category shall only be erected by a "Registered M .son".

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6. **JOINTS TO BLOCKWORK:**
All blockwork joints shall be neatly tooled with a 10mm rod jointer to form a neat concave recess to a good line, level and of consistent depth of approximately 6mm.

7. **WATERPROOFING:**
The Contractor shall ensure all blockwork above and below ground prevents the entry of water into the inside of the building. The Contractor shall provide warranties and prices for waterproofing from a number of manufacturers or their approved applicators and the type and price for the work shall be agreed. Internal strapping and lining as necessary to comply with insulation standards shall not proceed until evidence that the blockwork is free from leakage is demonstrated.

8. **WEEP HOLES:**
Provide weep holes in concrete blockwork at least 50mm below all bottom plates and below finished ground level, and below intermediate floor level, at approximately 800mm intervals. Drill or rake out weep holes to base of mortar bedding so as not to entrap any rain that might enter the walls.

BRICKLAYER

1. **PRELIMINARY AND GENERAL:**
Read and note all clauses under Preliminary and General of this specification where they apply to this trade.

2. **RELEVANT SPECIFICATIONS:**
All work shall be in accordance with the requirements of NZS 3604 and NZS 4229 as appropriate.

3. **MATERIALS:**
   (a) Bricks: Bricks for external veneers and foundation walls shall be of the colour and type selected and shall comply with NZS 368 Clay Building Bricks Grade A or B. All fair face brickwork shall be laid with their best face outwards.
   (b) Cement: Shall be ordinary Portland Cement and at the time of use shall comply with NZSS 3122 Portland Cement.
   (c) Plasticizers: Plasticizers shall be used in accordance with the manufacturer's instructions and no other additives are to be used in conjunction with these materials. On no account will further additions be made at the time of retempering mortars.
   (d) Water: Water shall be drinking quality.
   (e) Sand for Mortars: Sand used shall be Mercer No. 1 sand and/or shall comply with the relevant clauses of NZSS 3103, Sands for Mortar, Plasters and External Renderings.

4. **PREPARATION OF MORTAR:**
Mortar shall be prepared by mixing in an approved mixer. Measurement of materials shall be by volume using suitable containers. Mortar shall be mixed until a homogenous mass is obtained but for not less than 5 minutes. All mortar whether on boards or left in the mixer shall be used within 90 minutes. Mortar not used in this time shall be discarded.

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5. BRICKLAYING:
Bricks shall be laid in stretcher bond true to line level and plumb and in accordance with the best trade practice. All work shall be laid from the lowest corner and no corner shall be raised more than 900mm above wall line. Corners shall be racked back. On no account will toothing be permitted. All joints will be completely filled with mortar and the bricks shall be disturbed as little as possible after initial positioning. Joints shall, unless otherwise specified, be not more than 9.5mm thick and shall be tooled as directed as work proceeds. Where tapestry bricks are used care shall be taken to use a tool slightly smaller than joint width to prevent pushing mortar into the brick striations.

6. BRICK FOUNDATION WALLS:
Note where the following are shown on drawings and construct accordingly:-
(a) Reinforced clay brick masonry foundation walls not exceeding 2m in height, minimum width 130mm supporting single storey only; 190mm if supporting 2 storey or brick veneer and wall plate - all per NZS 3604.
(b) Reinforced clay brick masonry foundation walls exceeding 2m in height to be built by a "Registered Mason" to NZS 4229 or to Specific Design.
(c) Continuous single - wythe clay masonry foundation walls, supporting single storey only - strictly in accord with NZS 3604 with extra width of concrete beam and step detail for masonry veneer. Contractor to consult Owner on size and position of acess door.

7. MASONRY VENEER:
(a) Building Paper:
Run ordinary black bituminous breather type building paper to NZS 2295 (or fire resistant breather type) horizontally and well secure to outside face of framing from bearer to top plate. Repair tears and holes before constructing veneer.
(b) Clay Brick Veneer:
Construct brick veneer with approved face fixed ties at correct spacings in accordance with Appendix F of NZS 3604 and with materials and workmanship to NZS 4210 P.
A cavity of not less than 40mm and not more than 75mm shall be maintained between building paper and veneer. Where necessary weep holes shall be left every third joint for the discharge of water, as under Concrete Blocklayer. Keep the cavity and upstand clean, free from mortar protrusions and droppings. Keep pipes or electrical wire, junction boxes etc. out of the cavity. Build in the required number of sub-floor vents to provide not less than 3500mm² of ventilation per 1m² of floor area. Keep top of vents below bottom of bearers. Thoroughly clean down the face of work on completion and leave free of mortar stains and efflorescence.
(c) Vermin Proofing:
Secure galvanised wire mesh to bottom plates, slope downwards and across cavity and embed each length of mesh at least 25mm in mortar joints. Vermin proof dividing wall and wherever else necessary.
(d) Sills:
Sills to be brick on edge, brought to underside of wooden sills or aluminium joinery. Junction between to be filled with a suitable sealant as recommended by the Joinery Manufacturer and overpointed with cement mortar.

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(e) Split Stone or Concrete Block Veneer:
Cement composite bricks for external veneer and foundation walls shall be .....

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of the colour and type selected and shall be manufactured in accordance with NZS 3102 "Specification for Concrete Masonry Units". Construction shall accord with all earlier relevant clauses of this section.

8. CHIMNEY:
Chimney to be brick to Owner's preference. All reinforced to Territorial Authority requirements. Cast in C.I. tip grate to floor of hearth. Construct fireplace with jambs in 230mm work. Line with firebrick, with slanting sides and a tilted back giving a 75mm max. throat opening, finished 150mm above soffit of fireplace opening. Fill behind tilted back with mortar, with concave finish to top. Gather brickwork into flue liner and parge brickwork. Cast concrete lintel with 100mm bearing, reinforced with two 12mm diameter rods tied with 4mm wire at 150mm centres. Slope back face of lintel into throat. Construct flue with internal dimensions of 300mm x 300mm and build in 230mm diameter earthenware flue liner for full height, wrapped with heavy quality building paper.

FIREFRONT AND HEARTH:
Allow the P.C. Sum of $..... for selected firefront and hearth.